Sim2Real Hackathon
Installation Guide

Organised by: Robotech NITK | **Venue:** Hybrid

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1. Introduction

This guide explains how to install the required software, build the ROS 2 workspace, launch the Gazebo simulation, run the evaluator, and test your solution.

The simulation round uses:

- Ubuntu 24.04
- ROS 2 Jazzy
- Gazebo Harmonic
- RViz 2
- Python 3
- ROS-Gazebo bridge packages

Important: The official setup is intended for Ubuntu 24.04. Native Windows installation is not supported for this competition environment. If you are using Windows, use a compatible Ubuntu 24.04 installation or a properly configured virtual machine/dual-boot environment.

2. System Requirements

Before starting, update Ubuntu:

```
sudo apt update
sudo apt upgrade -y
```

Install basic development tools:

```
sudo apt install -y \git \
  python3-pip \
  python3-numpy \
  python3-opencv \
  python3-colcon-common-extensions \
  python3-rosdep \
  build-essential
```

3. Install ROS2 Jazzy

Follow the official ROS 2 Jazzy installation instructions:
https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html

Install the ROS 2 desktop package:

```
sudo apt update
sudo apt install -y ros-jazzy-desktop
```

Initialize rosdep if it has not already been initialized:

```
sudo rosdep init
rosdep update
```

If sudo rosdep init reports that it has already been initialized, continue with:

```
rosdep update
```

4. Install Simulation Dependencies

Install Gazebo Harmonic integration, camera processing, TF, and robot state publisher packages:

```
sudo apt install -y \
  ros-jazzy-ros-gz \
  ros-jazzy-cv-bridge \
  ros-jazzy-tf2-ros \
  ros-jazzy-robot-state-publisher \
  ros-jazzy-rviz2
```

Verify that ROS 2 is available:

```
source /opt/ros/jazzy/setup.bash
ros2 --version
```

5. Configure ROS2 Environment

To automatically source ROS 2 whenever a new terminal is opened, add the following line to your shell configuration:

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

Reload the configuration:

```
source ~/.bashrc
```

You can verify that the ROS environment is active by running:

```
printenv ROS_DISTRO
```

The output should be:

```
jazzy
```

6. Running the Simulation

The simulation requires three separate terminals:

- Simulation
- Evaluator
- Participant solution

In every terminal, source ROS 2 and the workspace:

```
source /opt/ros/jazzy/setup.bash
source ~/postman_ws/install/setup.bash
```

Terminal 1: Launch the simulation:

```
ros2 launch postman_pkg sim.launch.py tier:=1 seed:=42
```

Here:

- tier:=1 selects Tier 1.
- Use tier:=2 for Tier 2.
- Use tier:=3 for Tier 3.
- seed:=42 selects the procedural waypoint layout.

You may change the seed to test different layouts:

```
ros2 launch postman_pkg sim.launch.py tier:=3 seed:=100
```

* Official evaluation uses hidden seeds. Do not hardcode waypoint coordinates.

Terminal 2: Run the evaluator:

```
python3 src/postman_pkg/scripts/evaluator.py --tier 1
```

Run the evaluator with the same tier selected for the simulation.

Terminal 3: Launch the participant solution:

```
ros2 launch postman_solution solution.launch.py tier:=1
```

The launch file automatically starts the solution node for the selected tier.

7. Rebuilding After Changes

After modifying the participant code:

```
cd ~/postman_ws
source /opt/ros/jazzy/setup.bash
colcon build --symlink-install
source install/setup.bash
```

8. Useful links

How to install Ubuntu 24.04 on Windows (Virtualbox):
https://www.youtube.com/watch?v=DhVjgI57Ino

How to install ROS2 Jazzy:
https://www.youtube.com/watch?v=08o46x5SfJM

How to install Gazebo Harmonic:
https://www.youtube.com/watch?v=dY8JxldcuqA

9. Contact

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